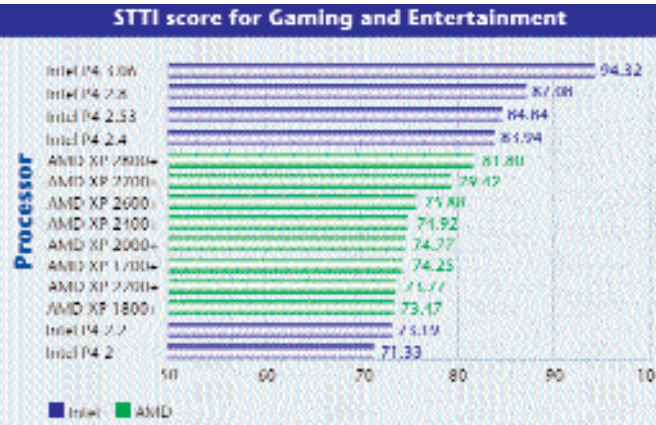




(arithmetic logic unit) and FPU (floating-point unit), to be exact. Since video encoding demands 100 per cent CPU utilisation, if the CPU fares well here, it can handle any other multimedia demands. To measure the performance of a processor during encoding, we encoded a standard MPEG-2 format video sequence using VirtualDub v1.4. The DivX codec used was version 4.01 and the setting for audio and video processing in VirtualDub was turned to full processing mode. The video compression bit rate was set at 900 Kbps.

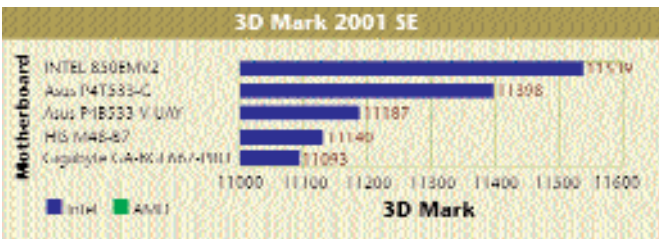
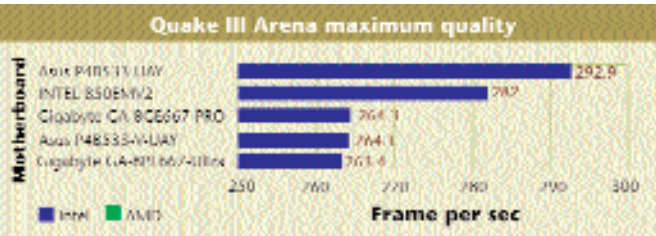
Entertainment is nothing without audio and nowadays, encoding audio is something that a PC user goes about doing very frequently. This also requires quite some amount of CPU power. The equation is simple: the faster the processor, the faster your audio will get encoded. To check just that, we used a wave file and encoded it into MP3 format and the time taken to get the job done was noted down.

The best motherboards for gaming and entertainment

Here we present to you some handpicked motherboards that stormed the gaming and entertainment tests and proved to be rock solid.

The best board to buy for stunning gaming experience is one that offers plenty of memory tweaks. Here the board that simply blew off any competition as far as overall appeal is concerned was the MSI 845MAX 2PE. Here is what it features: support for Bluetooth, RAID, FireWire, USB 2.0, six-channel audio, and support for PC 333 DDR-RAM. And this red monster sports HyperThreading, a feature that can do all the wonder you can possibly think of if coupled with a Hyper threaded processor.

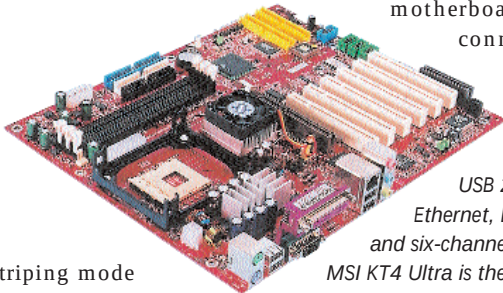
The board squashed the competition by its performance in the gaming arena. In *Quake III Arena* it squeezed 382 fps in the normal mode, which is very good considering the fact that we used a 266 MHz memory instead of 333 MHz. In the high-quality mode, it logged 368 fps and 249 fps in the max quality mode—this fall can be attributed to the slow memory that we



used, which caused a bottleneck. This is a very good example of what high-speed memory can do to the overall performance of the system.

The board logged a good score of 10933 in 3DMark 2001 SE, which by any standard is a good score for a board running a Ti4200 graphics card. The board gave an excellent score of 5338 in the PC Mark memory test—an excellent score. Compare this score with that of Intel's 850EMV2 board which runs on RDRAM memory running at 400 MHz. Although the gap between the two is notable, consider also that this board would perform much better when DDR 333 MHz memory sticks become more widespread. It thus becomes obvious that DDR memory is destined to give RDRAM the hiccups.

What's more, the MSI 845MAX 2PE's performance comes loaded with goodies one can drool over. With an onboard RAID controller hardwired onto the motherboard, you can connect two hard disks in the



With onboard USB 2.0, FireWire, Ethernet, Bluetooth, RAID and six-channel sound, the MSI KT4 Ultra is the most feature-rich board that one can buy

striping mode to further increase the disk subsystem speed to pull up some more fps when you play the latest games such as *DOOM III* that will demand every last bit from your system. To further enhance your gaming experience, the board features an onboard six-channel sound chip. Besides, you get four USB ports on the back panel. You also get FireWire ports so you use your PC for copying videos from your Camcorder. With an integrated Gigabit (1024 Mbps) LAN, playing games over your local network or over the Internet would be serious fun. The Northbridge adorns a nice silver heatsink-fan combo—a good feature and value addition by MSI. Apart from these things, support for Bluetooth by way of an optional Bluetooth kit further increases the scalability of the board. All these features, combined with a good performance, resulted in the board getting the highest STTI score of 81.63 in the field of Gaming and Entertainment.

But this wasn't the only board that caught our attention. Among the 12 boards that scored an STTI above 70, the platforms were dominated by Intel solutions with a split of eight and four between Intel and AMD. Close on the heels of the MSI board were two offerings from Gigabyte, the PRO and the ULTRA variant for Intel processors. Also, not too far behind were two solutions from Intel: the RDRAM supporting 850EMV2 and the DDR-based D845GEBV2. Separated by less than 3 points on the Gaming STTI Index, the cheaper